



L3 informatique

Course Notes
Software Engineering

Course 8
Deployment Diagram

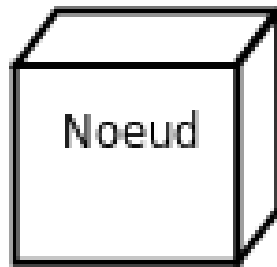
Presented by
Mr KHELIFA N.

Deployment Diagram

- **Specifies the system's hardware (physical) architecture.**
- **Describes the physical setup for the deployment of the software.**
- Identifies the hardware elements (PC, modem, workstation, server, etc.) and their physical arrangement (connections).
- **Describes the relationship between hardware and software.**
- Identifies the placement of executables (represented by artifacts) on these hardware elements.
- **Elements:** Node, Artifacts
Relationships: CommunicationPath, Deployment

Node

- A **node** is a computing resource on which artifacts can be deployed, via deployment relationships, to be executed.
- Nodes can have a complex internal structure defined by nesting and can be interconnected to represent specific situations.
- The internal structure of nodes can include only other nodes.
- Nodes can be connected to represent specific network topologies using connections.

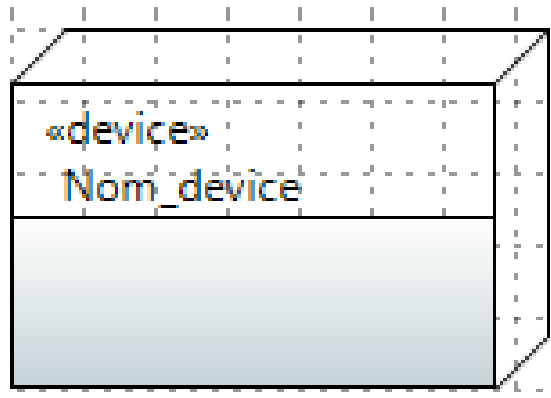


Node's type

- Nodes can be subtyped as:
 - **Devices** and
 - **ExecutionEnvironments.**

Device

- **Devices:**
- Represent the physical components of a machine.
- Are physical computing resources with processing capability on which artifacts can be deployed to be executed.
- Can be complex (i.e., they can consist of other devices), either through the namespace property or through attributes themselves typed by these devices.
- Examples: “application server,” “client workstation,” “mobile device,” and “embedded device.”

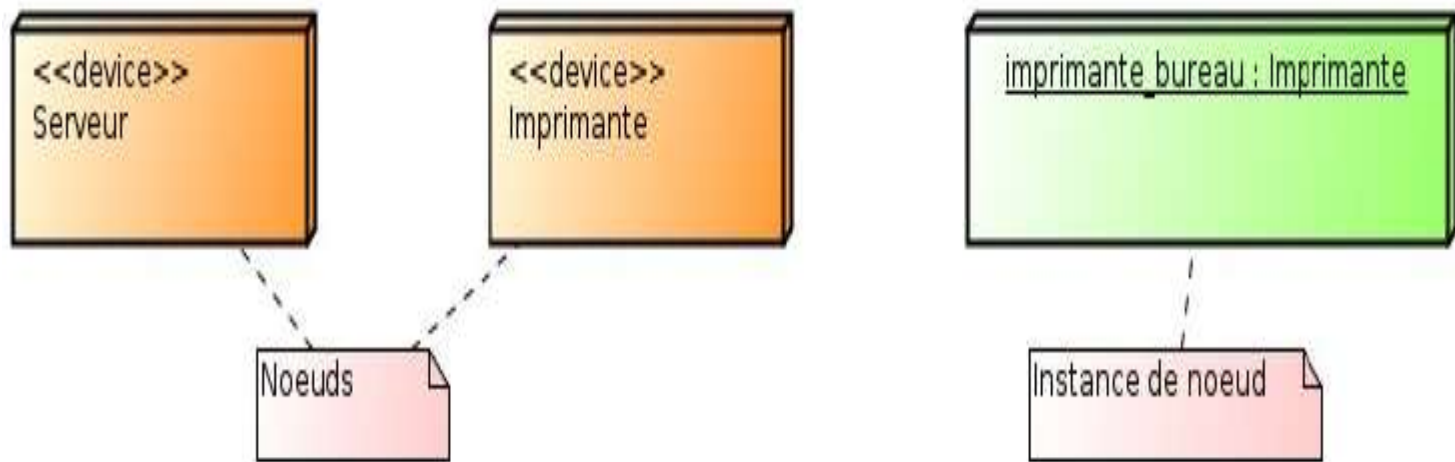


Example

- Two nodes, **server** and **printer**, are defined, along with an instance of a **printer**.

Exemple

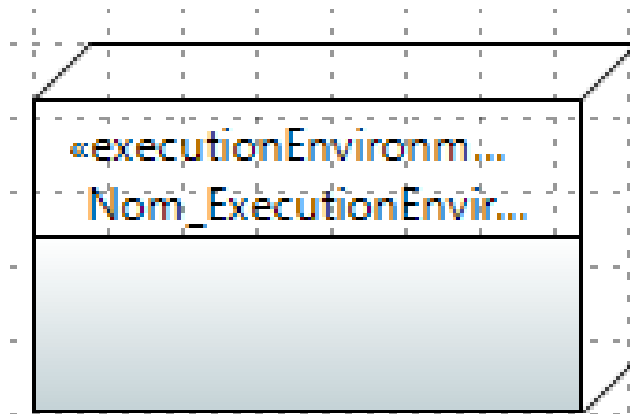
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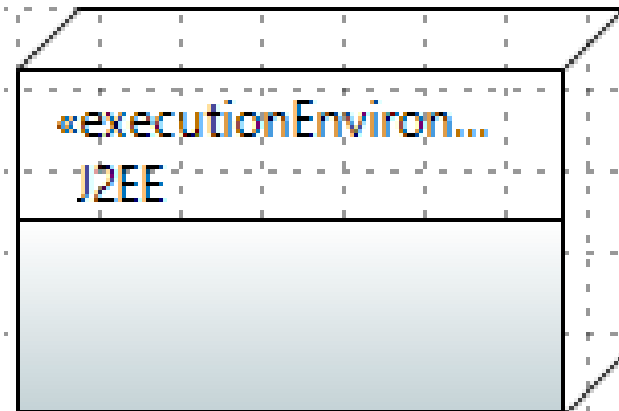
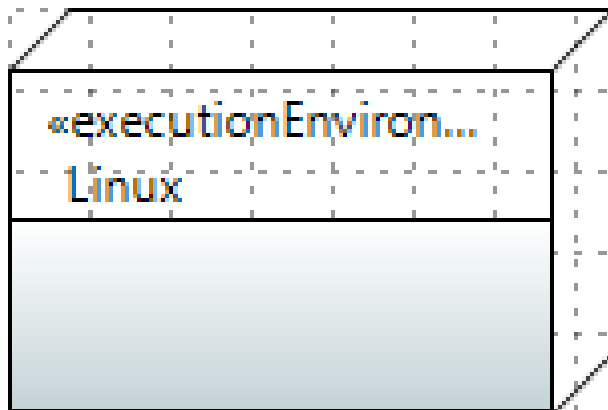
ExecutionEnvironments

- Represents standard software systems that application components may require at runtime.

Example: Operating systems (UNIX, ...), platforms (JEE, ...), ...

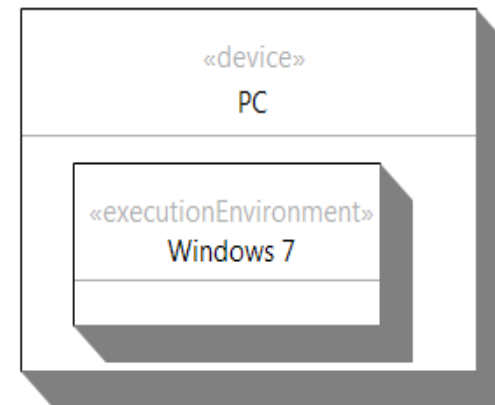
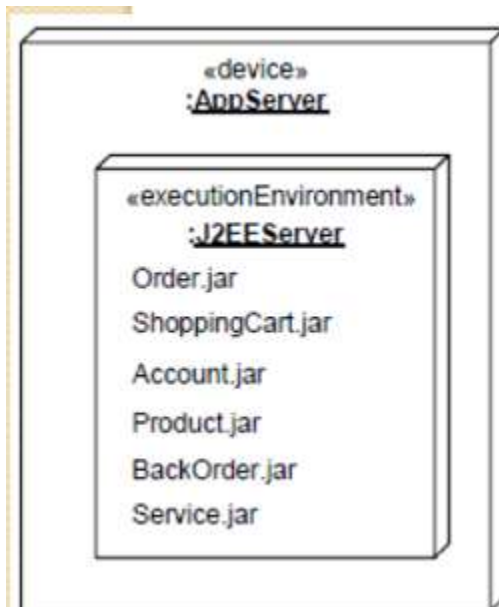


Example



Example

- In general, **ExecutionEnvironments** are assigned to a node.

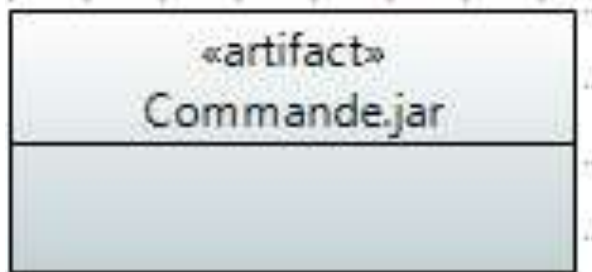


artifacts

- Represents an **information element** (generally reusable) that is used or produced by a software development process or by the operation of a system.
- **Examples of artifacts:**
 - Model files
 - Source files
 - Scripts
 - Executable files
 - Database tables
 - Deliverables for development
 - Word processing documents
 - Email messages

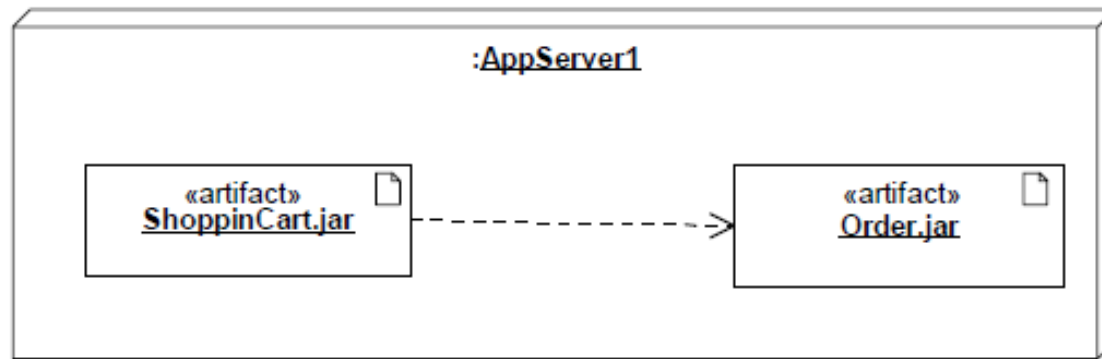
«artifact»
NameArtifact

Example

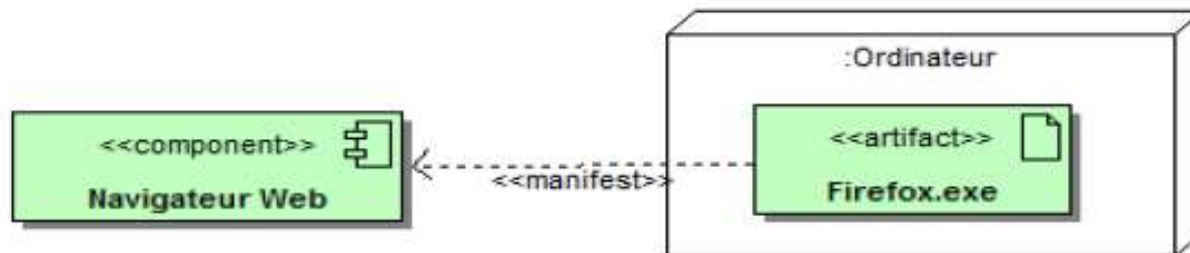


Relationships between artifacts and other elements

- **Dependency:** between artifacts

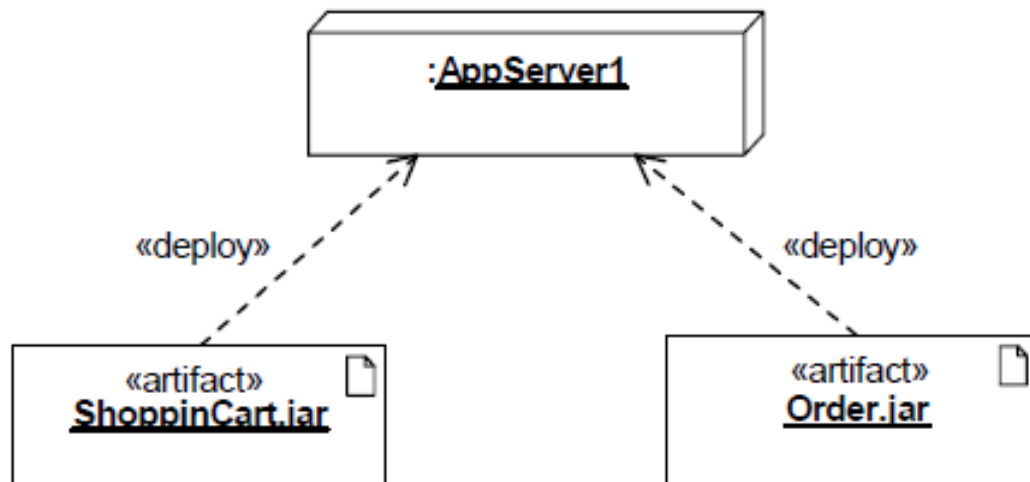


- **Manifestation:** between an artifact and components



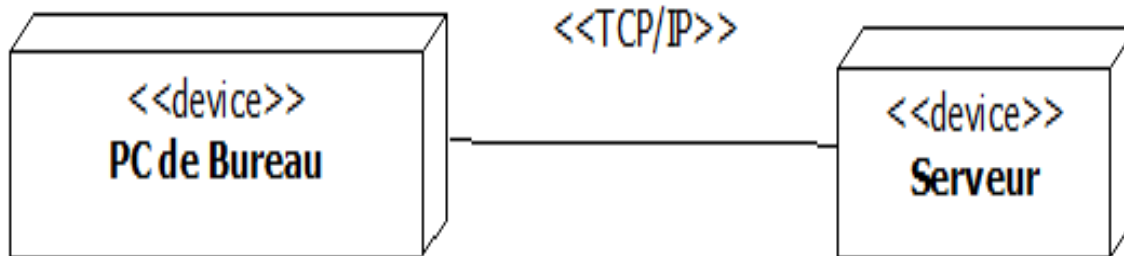
Relationships between artifacts and other elements

- **Deployment:** between an artifact and a node



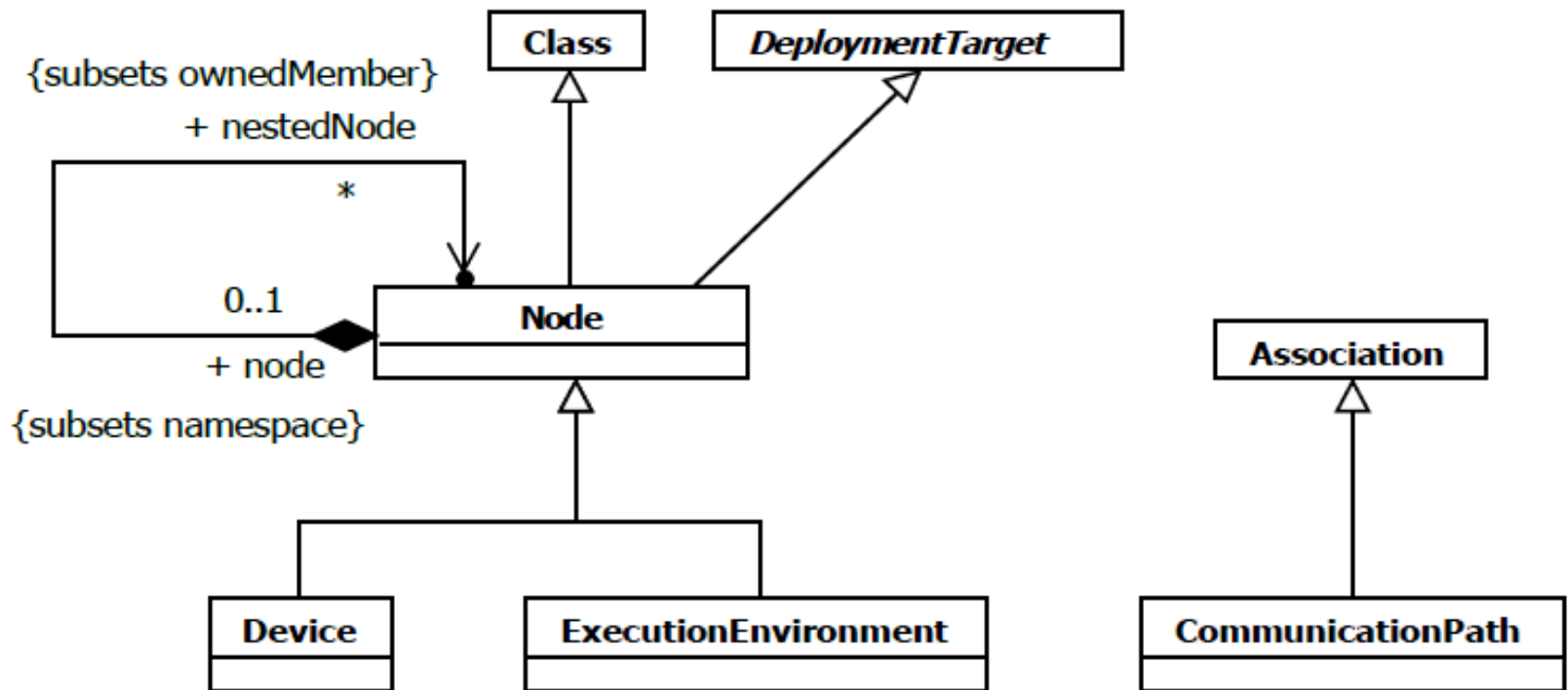
Communication Path

- Specifies communication paths between nodes that enable the exchange of information (messages and signals).



The nature of the association is generally complemented by a **stereotype** to describe the communication protocol used (e.g., RS232, TCP/IP, bus).

Abstract syntax

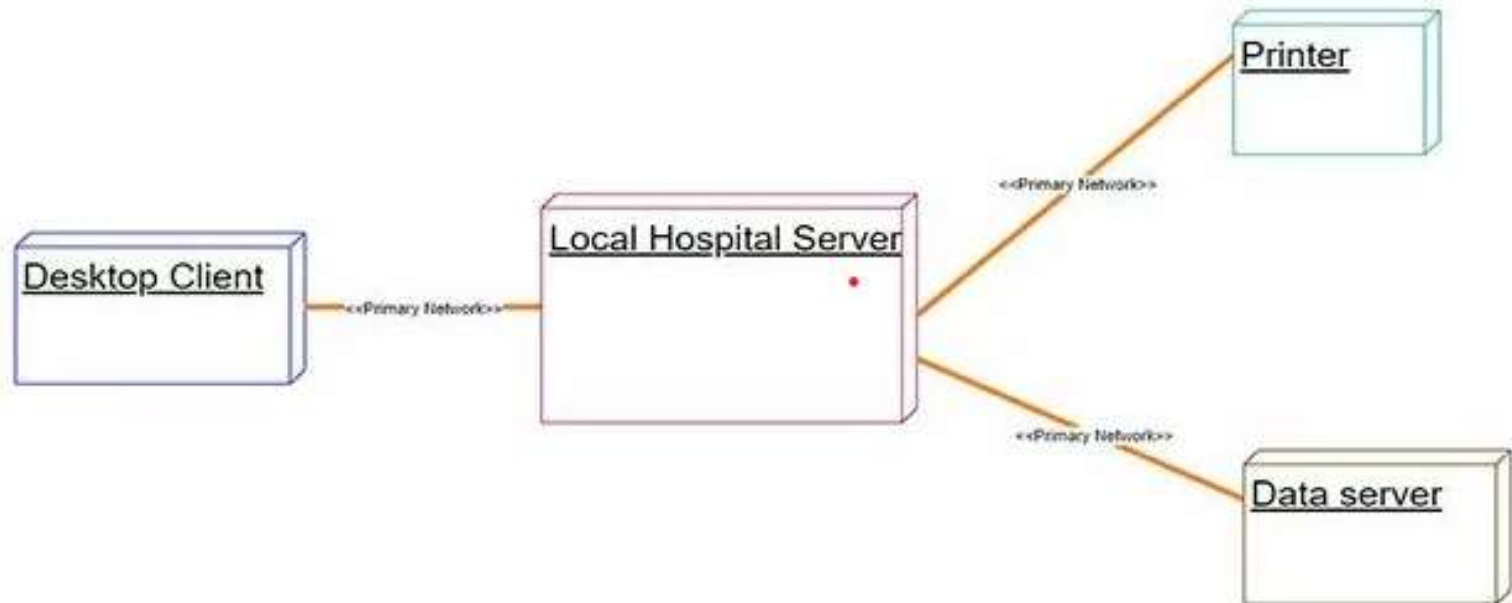


Example 1

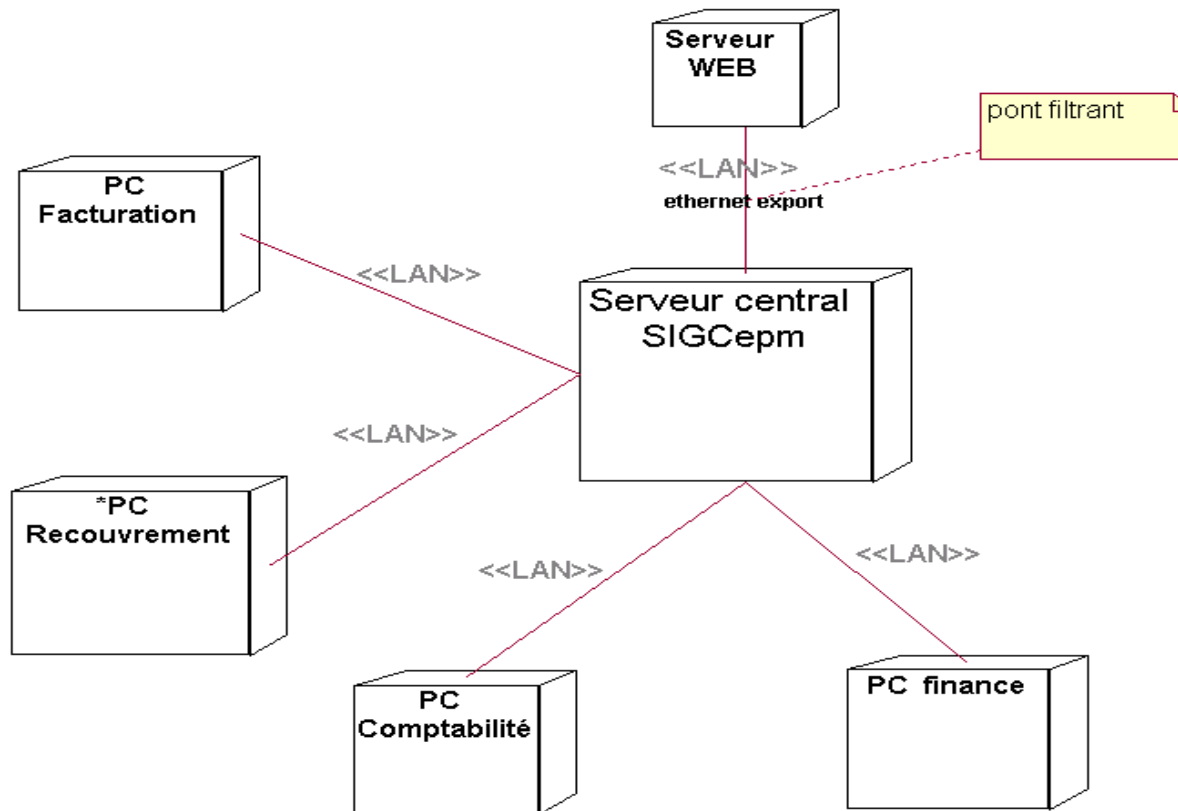
- A client PC is connected to a local hospital server, which is itself connected to a printer and a database server.

Example I

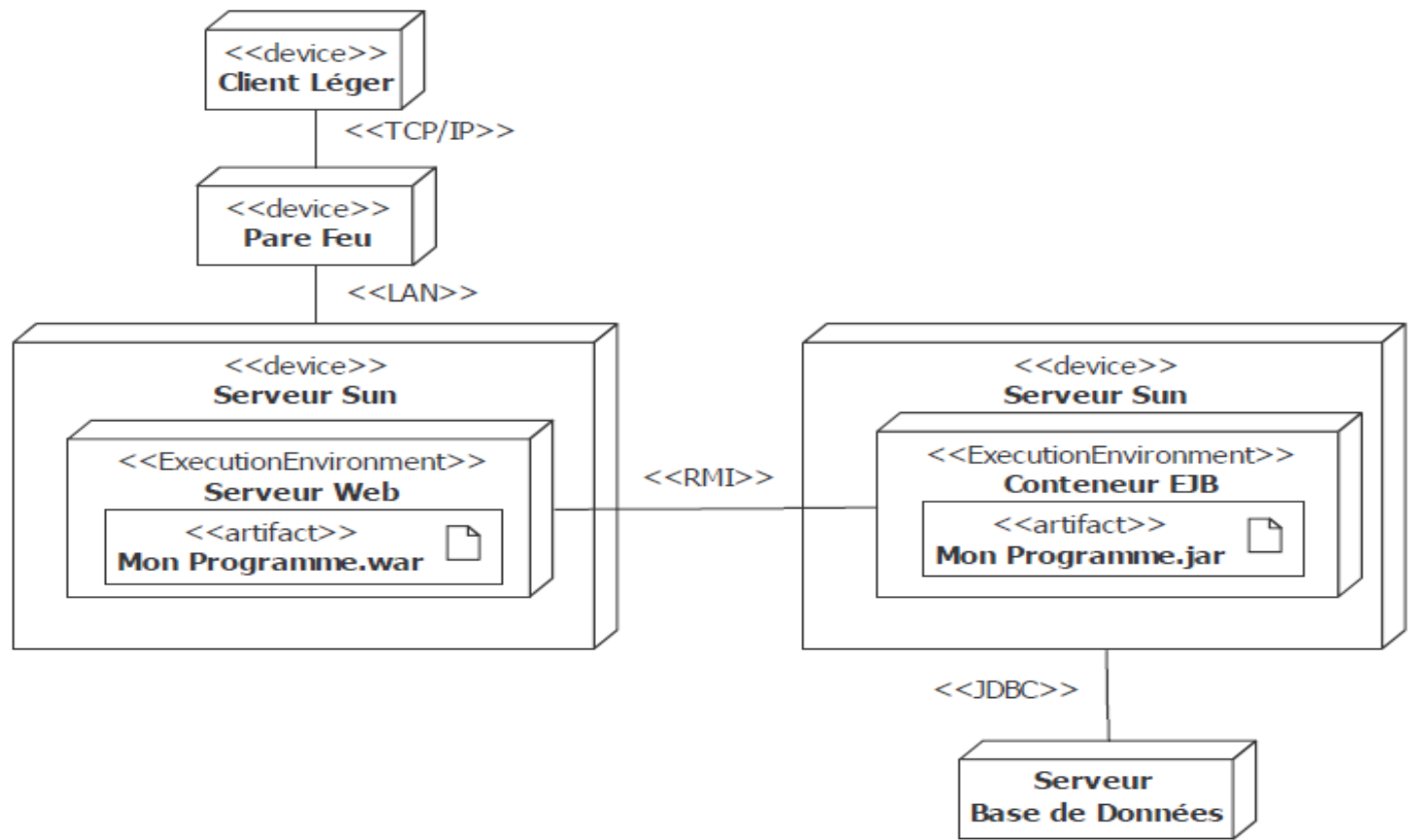
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Exemple (I)



Exemple





Merci de votre attention

Les diagrammes UML 2.5.1

